CSE 641 CV Project Report Week-6

Project Name: Athletic Readiness: Data Generation

Group Name: Deep Vision

Objective

The feature of this week was to use Generative Adversarial Network (WGAN-GP) on the same preprocessed Week 5 week data and assess the quality of the produced synthetic data. Our group was concerned with determining whether the generated data is distributed and correlated with the data initially and comparing with the VAE in the Week 5.

Data Collection

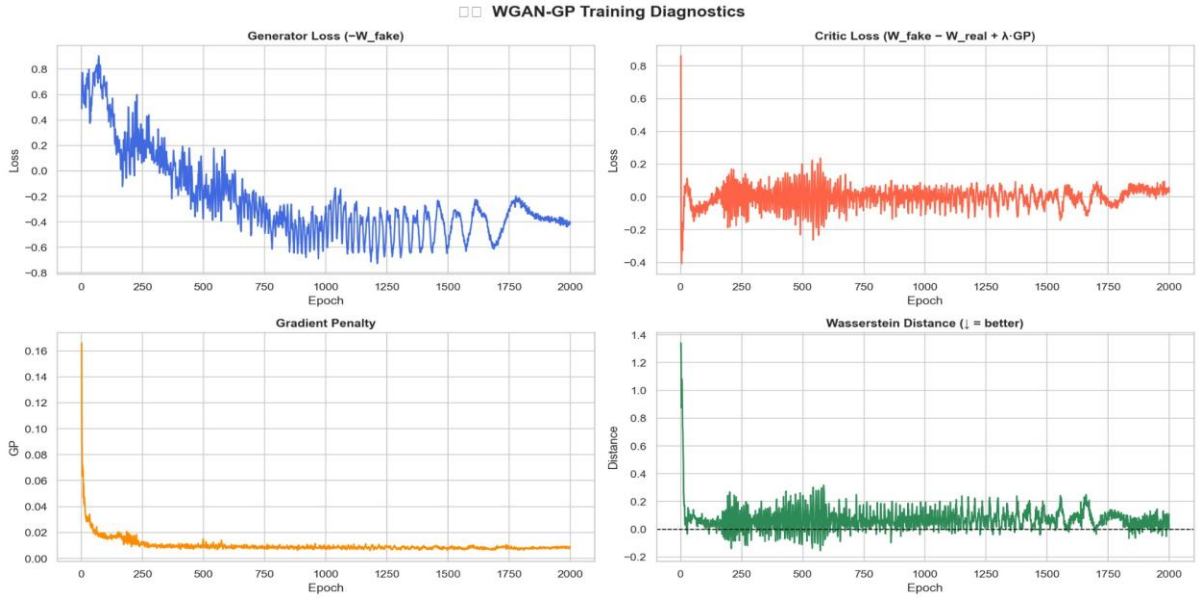
The training of the VAE was done using the same dataset as that of the WGAN-GP. It contains 11 feature column and 524 records that had been preprocessed and cleaned in Week 4. The two generative models should be compared using the same training data that would allow a fair comparison between the two.

Methodology: WGAN-GP Implementation

The WGAN-GP model was fed with the data that was capable of learning the underlying data distribution and could be used to produce samples. The input random noise (latent dim = 8) is transformed to synthetic player records with the help of a Generator and then a Critic evaluates the Wasserstein distance of the real and generated distributions. A gradient penalty is a training stability requirement which enforces the Lipschitz constraint.

To make a fair evaluation assessment, the number of records that it produced was 600 synthetic player records after 2,000 training epochs, and this was equal to the number of VAE records.

Training Diagnostics

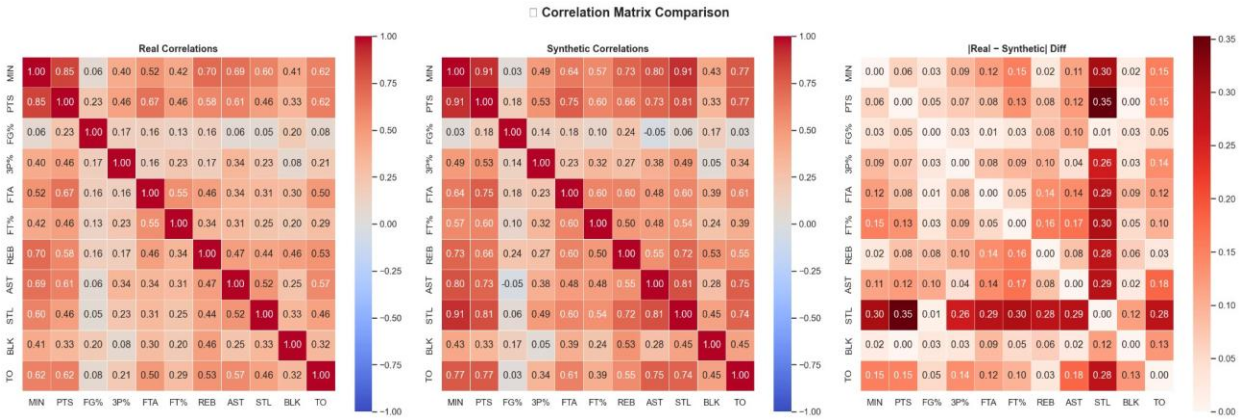


The Gradient Penalty rapidly stabilized at a Gradient at approximately 0.01 indicating that it was successfully enforced. The Wasserstein Distance dropped to almost zero, which means that the generated distribution was becoming increasingly similar to the actual data.

Evaluation Generated Data

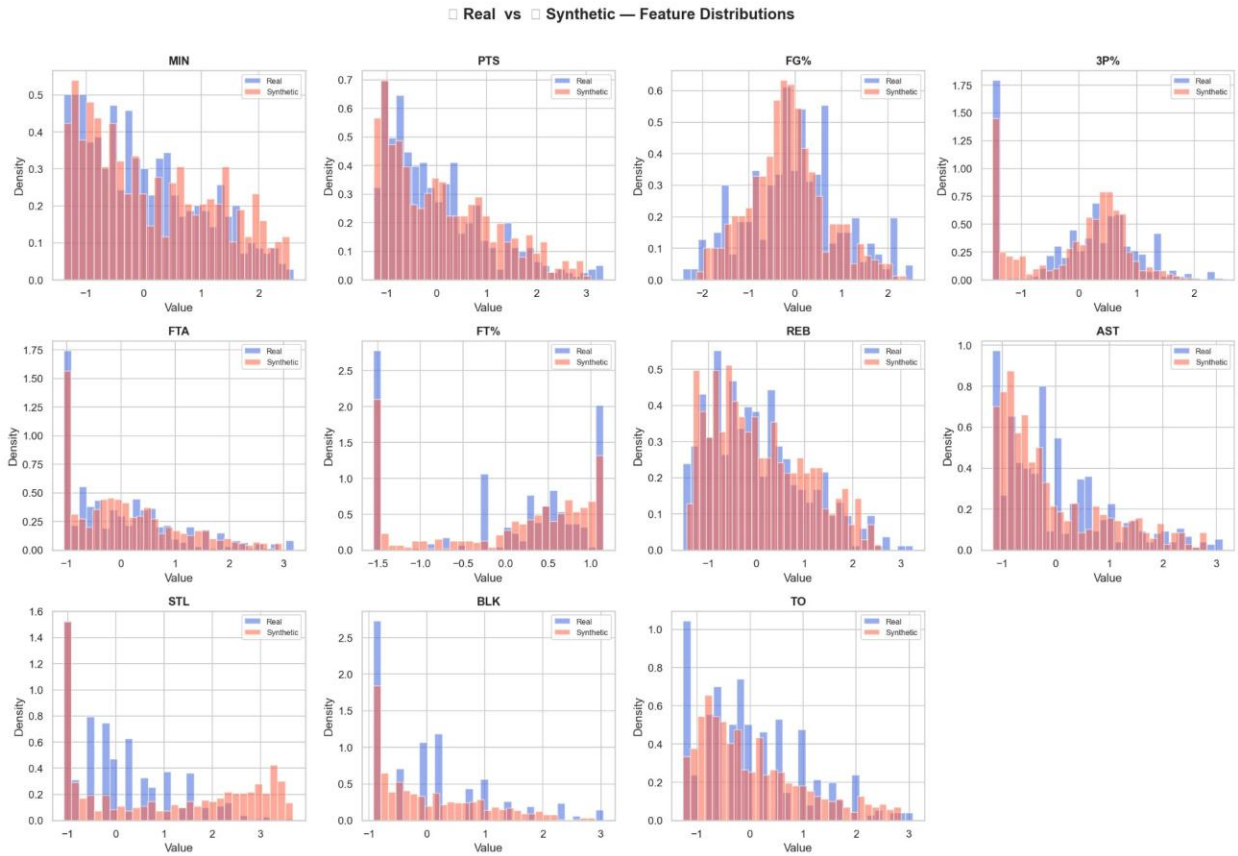
- Correlation Analysis

A heatmap of correlation was plotted with the real data and the synthetic data. The pattern of correlation among features was similar as indicated by the heatmaps. Certain pair-wise correlations of STL exhibited a marginally greater difference (up to 0.35) of the VAE outcome, however, the general inter-feature dependency structure was likewise maintained.

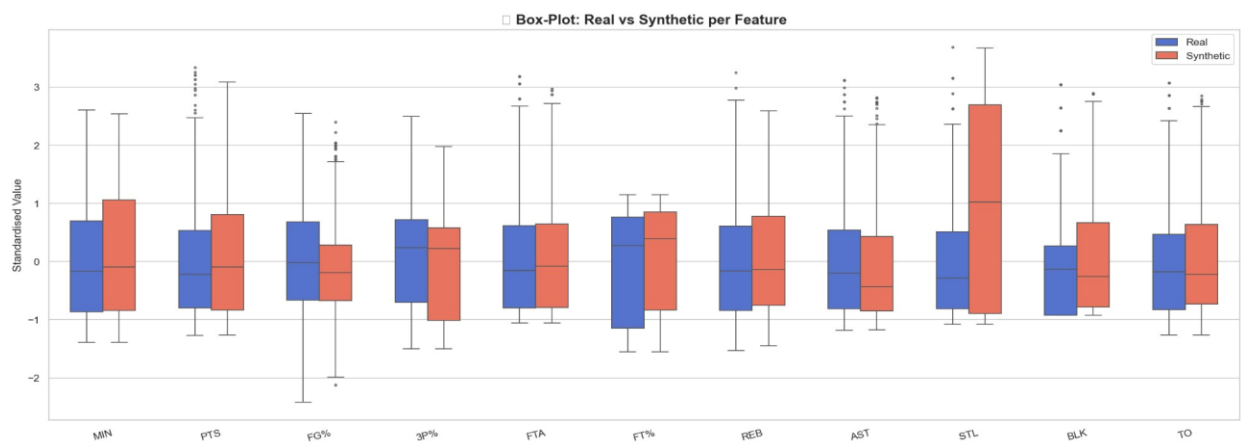


- Feature Distribution Analysis

The real and generated features were plotted in histograms of each of the 11 features. Most features (MIN, PTS, FTA, REB, AST, TO) had distributions that were close to each other. The veil of sparsity, i.e. STL and BLK, was more divergent than the VAE, which happens to be one of the challenges of GANs on skewed tabular input.

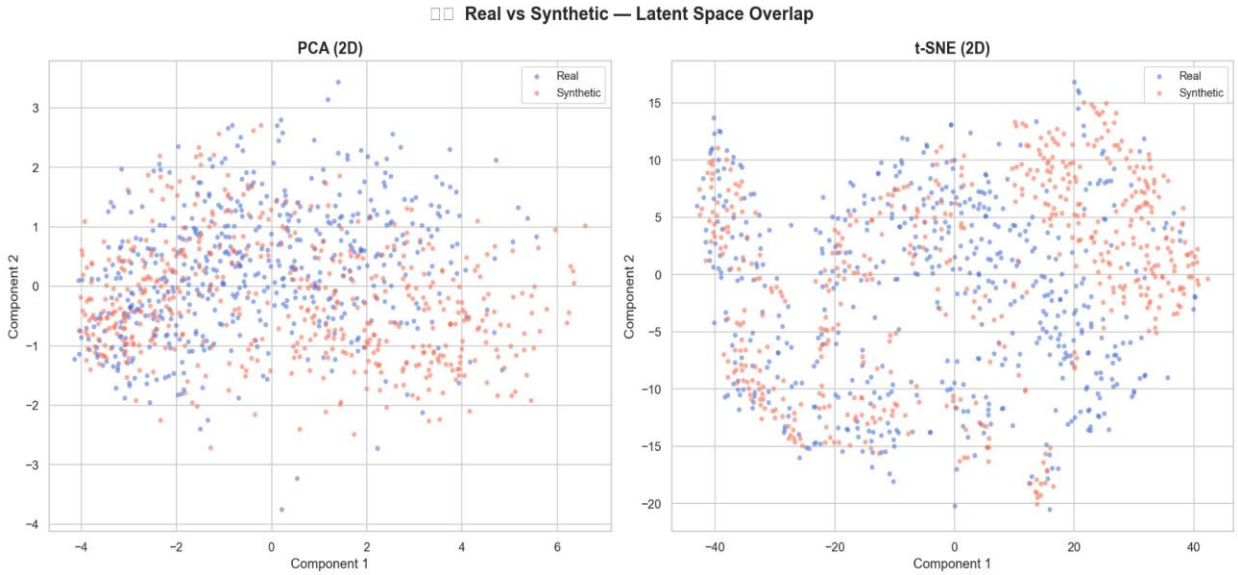


- Box-Plot Comparison



- Latent Space Overlap

The accessibility of synthetic data in the same latent space, as the real data, is confirmed by PCA and t-SNE visualisations, and there are high quality global overlap in both projections.



Results & Observations

- This is because the WGAN-GP was able to generate synthetic data that is widely similar to original data.
- Training stability 2,000 epochs: This was training that trained the Wasserstein Distance to values close to 0.
- Most of the features had maintained their feature-wise distributions and correlation structure.
- Sparse features (STL, BLK, FT%) were more divergent than the VAE results at Week 5.
- PCA and t-SNE prove that they have good coverage of the latent space with small boundary gaps.

Conclusion

The WGAN-GP model has been identified to be useful in the production of quality synthetic information. The ex post facto and the generated data sets show quite similar distributions and correlation, which proves the power of the approach. In comparison with VAE, the GAN has less unstable adversarial training and a reduced divergence with sparse features. This marks the end of the implementation step, and opens up the groundwork to an official VAE vs GAN comparison.

Plan for Next Week

- Implement testing strategies for synthetic data validation:
  TRTR (Train Real, Test Real)
  TSTR (Train Synthetic, Test Real)
  TRTS (Train Real, Test Synthetic)
  TSTS (Train Synthetic, Test Synthetic)
- Train a classification/regression model on the real dataset and evaluate baseline performance.
- Train the same model using synthetic data generated by VAE and GAN.
- Comparing the model performance will be done with all the four testing strategies.